

Table 1: Replicate holdout, 12L12D only: RMSE (mm, mean $\pm$ sample SD) / relative RMSE (% , mean) across three seeds, except the single-fit logistic ODE. Targets are split-specific replicate means. Bold marks the lowest unrounded mean in each split. Only the two added PhytoODE variants use binary illumination as an input; both retain ordinary logistic physics. Fixed settings reuse the selected no-light PhytoODE hyperparameters. The tuned variant has an additional search budget. The matched latent ODE uses the no-light PhytoODE settings with both physics coefficients zero.

Model	Train	Validation	Test
PhytoODE + light (tuned)	$0.190 \pm 0.008/3.75$	$0.278 \pm 0.001/5.48$	$0.346 \pm 0.008/6.85$
PhytoODE + light (fixed)	$0.171 \pm 0.011/3.38$	$0.289 \pm 0.007/5.70$	$0.329 \pm 0.006/6.52$
PhytoODE	$0.191 \pm 0.002/3.76$	$0.293 \pm 0.001/5.78$	$0.365 \pm 0.008/7.22$
Latent ODE	$0.190 \pm 0.007/3.75$	$0.297 \pm 0.003/5.86$	$0.368 \pm 0.007/7.28$
Logistic-PINN	$0.219 \pm 0.013/4.32$	$0.304 \pm 0.002/6.00$	$0.361 \pm 0.014/7.14$
LSTM-NN	$0.137 \pm 0.029/2.71$	$0.288 \pm 0.014/5.68$	$0.361 \pm 0.007/7.15$
Random forest	$0.364 \pm 0.013/7.17$	$0.472 \pm 0.027/9.33$	$0.452 \pm 0.014/8.95$
Logistic ODE	$0.370/7.29$	$0.408/8.07$	$0.485/9.61$
Latent ODE (matched)	$0.190 \pm 0.007/3.75$	$0.297 \pm 0.003/5.86$	$0.368 \pm 0.007/7.28$

Table 2: 36/60-h holdout, 12L12D only: RMSE (mm, mean $\pm$ sample SD) / relative RMSE (% , mean) across three seeds, except the single-fit logistic ODE. Targets are split-specific replicate means. Bold marks the lowest unrounded mean in each split. Only the two added PhytoODE variants use binary illumination as an input; both retain ordinary logistic physics. Fixed settings reuse the selected no-light PhytoODE hyperparameters. The tuned variant has an additional search budget. The matched latent ODE uses the no-light PhytoODE settings with both physics coefficients zero.

Model	Train	Validation	Test
PhytoODE + light (tuned)	$0.131 \pm 0.014/2.87$	$0.267 \pm 0.001/5.87$	$0.266 \pm 0.026/4.22$
PhytoODE + light (fixed)	$0.138 \pm 0.033/3.03$	$0.275 \pm 0.012/6.05$	$0.241 \pm 0.011/3.83$
PhytoODE	$0.143 \pm 0.033/3.14$	$0.268 \pm 0.006/5.90$	$0.442 \pm 0.086/7.01$
Latent ODE	$0.146 \pm 0.005/3.19$	$0.282 \pm 0.015/6.21$	$0.411 \pm 0.049/6.53$
Logistic-PINN	$0.114 \pm 0.016/2.48$	$0.289 \pm 0.012/6.35$	$0.381 \pm 0.020/6.05$
LSTM-NN	$0.058 \pm 0.046/1.26$	$0.292 \pm 0.008/6.43$	$0.503 \pm 0.087/7.97$
Random forest	$0.640 \pm 0.026/14.00$	$0.626 \pm 0.038/13.78$	$1.322 \pm 0.025/20.99$
Logistic ODE	$0.367/8.03$	$0.451/9.92$	$0.311/4.94$
Latent ODE (matched)	$0.100 \pm 0.023/2.19$	$0.291 \pm 0.013/6.40$	$0.448 \pm 0.008/7.10$